

Heptadecagon

Gauss's 17-gon

Only a compass and a straightedge. Every point is marked on the front.

1. Quarter an angle

Bisect OB to find its middle, H . Bisect OH to find I , a quarter of the way up. Bisect angle OIA , then bisect the half next to IO . Where the line meets line OA : E .

2. 45 degrees, then a circle

At I , make a right angle to IE . Mark two points on line IE , one on each side of I , the same distance away. From each, draw a wider arc. Join I to where the arcs cross. Bisect the right angle on O 's side of IE to get 45 degrees. Where it meets line OA : F . Find the middle of AF : equal arcs from A and from F , then join their crossings. The middle is M . Compass on M , opened to A : draw the circle. It meets OB at K .

3. One more circle

Compass on E , opened to K : draw the circle. It meets line OA at N_3 and N_5 . Raise a perpendicular at each, up to the big circle: P_3 and P_5 . N_3 and M are only 0.74 mm apart, so their pits look like one.

P_3 is exactly $3/17$ of the way round from A . P_5 is exactly $5/17$.

4. Walk it round

Open the compass from P_3 to P_5 and step it from A toward P_3 : you land $2/17$ of the way round. From there to P_3 is one side, exactly $1/17$. Walk that width round 17 times, checking each mark against its tick, and join the marks.

Inspired by Numberphile:
Heptadecagon (17-gon)

